
Optimizing Pedestrian Crossings and Traffic Signal Control for Urban Mobility

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Abstract

This research aims to analyze the impact of pedestrian crossing timings on vehicular traffic flow in urban environments and propose an adaptive traffic signal control algorithm using Q-learning. The operational and safety performance of the proposed algorithm was tested using the Simulation of Urban Mobility (SUMO). The results demonstrate significant improvements in traffic flow and safety, especially under high Market Penetration Rates (MPRs) of Connected and Automated Vehicles (CAVs).

Keywords: Urban Traffic, Q-learning, Traffic Signal Control, SUMO, Adaptive Algorithms, Connected and Automated Vehicles (CAVs)

1 Introduction

Urban traffic congestion, especially during large events like the 2020 Tokyo Olympics, highlights the need for more efficient traffic management solutions. Enhancing road safety and promoting efficient urban mobility by reducing idling times at traffic lights are critical goals of this research.

Traffic signal control is a crucial element in managing urban traffic flow. Traditional methods, such as Fixed-Time (FT) control, rely on pre-set timers that do not adapt to real-time traffic conditions, leading to inefficiencies. Adaptive Traffic Control Systems (ATCS), such as SCATS and SCOOT, adjust signal timings based on real-time data from sensors but require substantial infrastructure. Recent advances in machine learning, particularly reinforcement learning algorithms like Q-learning, offer new possibilities for optimizing traffic signal control by learning from real-time traffic conditions.

This study focuses on developing a Q-learning-based adaptive traffic signal control al-

gorithm that dynamically adjusts signal timings to improve both vehicular traffic flow and pedestrian safety at urban intersections. The integration of Connected and Automated Vehicles (CAVs) data enhances the algorithm's accuracy and responsiveness, making it a promising solution for future smart city applications.

2 Methodology

The methodology section outlines the design and implementation of the proposed Q-learning-based traffic signal control algorithm.

2.1 State Representation

In the Q-learning algorithm, states are defined based on the queue lengths of each approach at the intersection. The state representation is crucial for the algorithm to understand the traffic conditions and make informed decisions. Table 1 presents the state set used in this study.

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Table 1: State set for the proposed algorithm.

State	Link Order
S1	$q1 \geq q2 \geq q3 \geq q4$
S2	$q1 \geq q2 \geq q4 \geq q3$
S3	$q1 \geq q3 \geq q2 \geq q4$
S4	$q1 \geq q4 \geq q2 \geq q3$
S5	$q1 \geq q3 \geq q4 \geq q2$
S6	$q1 \geq q4 \geq q3 \geq q2$
S7	$q2 \geq q1 \geq q3 \geq q4$
S8	$q2 \geq q1 \geq q4 \geq q3$
S9	$q3 \geq q1 \geq q2 \geq q4$
S10	$q4 \geq q1 \geq q2 \geq q3$
S11	$q3 \geq q1 \geq q4 \geq q2$
S12	$q4 \geq q1 \geq q3 \geq q2$
S13	$q2 \geq q3 \geq q1 \geq q4$
S14	$q2 \geq q4 \geq q1 \geq q3$
S15	$q3 \geq q2 \geq q1 \geq q4$
S16	$q4 \geq q2 \geq q1 \geq q3$
S17	$q3 \geq q4 \geq q1 \geq q2$
S18	$q4 \geq q3 \geq q1 \geq q2$
S19	$q2 \geq q3 \geq q4 \geq q1$
S20	$q2 \geq q4 \geq q3 \geq q1$
S21	$q3 \geq q2 \geq q4 \geq q1$
S22	$q4 \geq q2 \geq q3 \geq q1$
S23	$q3 \geq q4 \geq q2 \geq q1$
S24	$q4 \geq q3 \geq q2 \geq q1$

2.2 Action Representation

Actions in the proposed algorithm involve adjusting signal timings based on current traffic conditions. Two primary actions can be chosen: extending the green time by 1 second or not extending it. These actions allow the algorithm to dynamically respond to real-time traffic conditions and optimize traffic flow.

The action space is represented as follows:

- **Extend Green Time:** Increase the green signal duration for the current phase by 1 second.
- **Do Not Extend Green Time:** Maintain the current green signal duration.

2.3 Reward Function

The reward function is designed to encourage actions that reduce traffic congestion and im-

prove safety. It integrates data from loop detectors and CAVs to improve accuracy. The queue length detected by both methods is used to calculate the reward, which adjusts the weight of the two methods through MPR.

The reward function $R(s, a)$ is defined as follows:

$$R(s, a) = (1 - \epsilon)q + \epsilon q' \quad (1)$$

where:

- q is the queue length estimated by loop detectors.
- q' is the queue length detected by CAVs.
- ϵ is the current Market Penetration Rate (MPR) of CAVs.

This weighted approach allows the algorithm to balance the accuracy of data from different sources and make informed decisions that optimize traffic flow and safety.

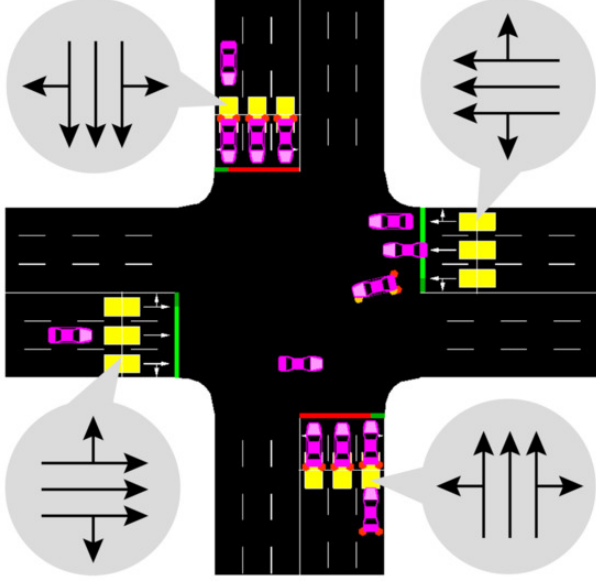


Figure 1: Sumo intersection simulation.

3 Results

3.1 Simulation Settings

3.1.1 Intersection Geometry

The simulations were conducted in SUMO using a virtual intersection with four legs and three lanes per approach. Each lane is 250 meters long, and loop detectors are placed at the beginning and end of each lane.

3.1.2 Traffic Demand

A balanced traffic demand of 1000 vehicles per hour (veh/h) was set for each approach. The simulation also included various MPRs ranging from 0% to 100%.

3.1.3 Vehicle Types

The simulation included 11 MPRs to represent the proportion of CAVs in the traffic flow. The Intelligent Driver Model (IDM) and Cooperative Adaptive Cruise Control (CACC) model were used for Human-Driven Vehicles (HDVs) and CAVs, respectively.

3.2 Simulation Results

3.2.1 Operational Performance

The operational performance was measured by the average delay experienced by vehicles. Delay is defined as the difference between the current travel time since a vehicle enters the incoming lane of the intersection and the expected minimum travel time of the lane. The proposed algorithm reduced average delays by 9.1% compared to FT and TAC.

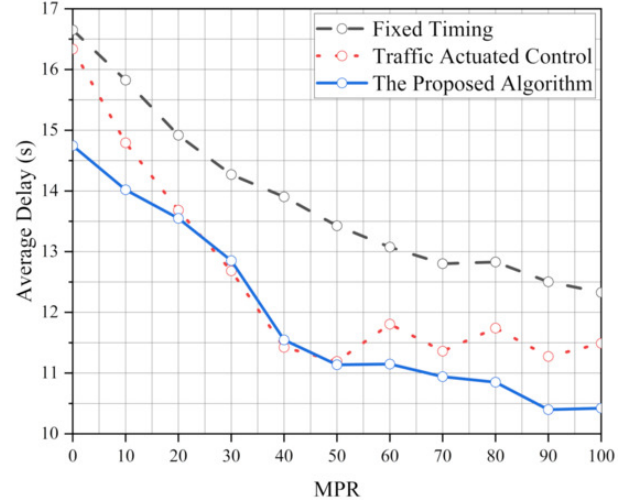


Figure 2: The average delay under various MPRs using the proposed algorithm, FT, and TAC in 1000 veh/h.

Figure 2 shows the average delay under various MPRs using FT, TAC, and the proposed algorithm. It can be clearly found that the operational performances of TAC and the proposed algorithm are better than FT. In the numerical experiment, the average delay of the proposed algorithm reduces by 9.1–16.9% than FT. The optimized performance is significant under high MPRs (more than 50%).

3.2.2 Safety Performance

The safety performance was estimated by the conflict rate using Time-to-Collision (TTC). The proposed algorithm showed significant reductions in conflict rates compared to FT and TAC, particularly under high MPRs.

4 Discussion

4.1 Q-Learning Algorithm

The Q-learning algorithm used in this study follows a reinforcement learning approach, where the algorithm learns to optimize traffic signal timings through trial and error. The key steps of the algorithm are as follows:

- **Observe Current State:** Analyze the current traffic conditions at the intersection.
- **Choose an Action:** Select an action (e.g., change the traffic signal) based on an exploration-exploitation strategy.
- **Execute the Action:** Implement the chosen action and observe the result.
- **Update Q-Value:** Update the Q-value based on the observed reward. If the result is positive (reduced congestion), the Q-value is increased. If the result is negative (increased congestion), the Q-value is adjusted accordingly.
- **Repeat until Convergence:** Continue this process until the Q-values stabilize, indicating that the algorithm has learned the optimal signal timings.

The Q-learning formula used in this study is:

$$Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_a Q(s', a') - Q(s, a)] \quad (2)$$

where:

- $Q(s, a)$ is the current Q-value for state s and action a .
- α is the learning rate.
- r is the reward.
- γ is the discount factor.
- $\max_a Q(s', a')$ is the maximum estimated future reward for the next state s' .

4.2 Robust Analysis

The robustness of the proposed algorithm was tested by increasing traffic demand to simulate peak hours. The results indicated that the proposed algorithm outperformed FT and TAC even under higher traffic demands.

5 Conclusions

The proposed Q-learning-based adaptive traffic signal control algorithm significantly improves the operational and safety performance of intersections, particularly under high MPRs. The algorithm provides a robust and effective solution for managing traffic flow in urban environments, offering a foundation for future smart city traffic management systems.

Future work will focus on further refining the algorithm and testing it in more complex urban environments. Integrating additional data sources and enhancing the learning mechanism are potential areas for improvement.

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